

2026 A-level

Q.7

(a) - *Sector Area Equation*

$$\begin{aligned}
 R_1 &= 2R_2 \\
 \pi r^2 \frac{\theta}{2\pi} - \frac{1}{2} r^2 \sin \theta &= 2 \left[\pi r^2 \frac{\pi - \theta}{2\pi} - \frac{1}{2} r^2 \sin(\pi - \theta) \right] \\
 \frac{\theta}{2} - \frac{1}{2} \sin \theta &= \pi - \theta - \frac{1}{2} \sin \theta \\
 \frac{\sin \theta}{2} &= \pi - \frac{3\theta}{2}
 \end{aligned}$$

Q.8

(a) **Integral:** $\int_0^3 x e^{2x} dx$

(b)

$$\begin{aligned}
 \int_0^3 x e^{2x} dx &= \left[\frac{x e^{2x}}{2} \right]_0^3 - \int_0^3 \frac{e^{2x}}{2} dx \\
 &= \frac{3e^6}{2} - \left(\frac{e^6}{4} - \frac{1}{4} \right) \\
 &= e^6 + \frac{1}{2}
 \end{aligned}$$

Q.9

(a) - *Trigonometric Identities*

$$\begin{aligned}
 (\cos \theta + \sin \theta) \left(\frac{1}{\sin \theta} - \frac{1}{\cos \theta} \right) &= (\cos \theta + \sin \theta) \frac{(\cos \theta - \sin \theta)}{\sin \theta \cos \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \\
 &= \frac{1 - \tan^2 \theta}{\tan \theta} \\
 &= 2 \times \frac{1 - \tan^2 \theta}{2 \tan \theta} \\
 &= 2 \cot 2\theta
 \end{aligned}$$

(b)

$$\begin{aligned}
 2 \cot 2\theta &= \frac{4}{5 \sin^2 2\theta} \\
 2 \sin 2\theta \cot 2\theta &= \frac{4}{5} \\
 \sin 4\theta &= \frac{4}{5} \\
 4\theta &= \dots\dots\dots - 360^\circ < x < 360^\circ
 \end{aligned}$$

Q.10**(a)** - Differentiation, slope at $x = 2$

$$\begin{aligned}
 y &= \frac{7x}{\sqrt{3x^2 + 4}} \\
 \frac{dy}{dx} &= \frac{7\sqrt{3x^2 + 4} - 7x \cdot \frac{1}{2}(3x^2 + 4)^{-\frac{1}{2}}(6x)}{3x^2 + 4} \\
 &= \frac{7}{\sqrt{3x^2 + 4}} - \frac{21x^2}{(3x^2 + 4)^{\frac{3}{2}}}
 \end{aligned}$$

$$\begin{aligned}
 \left. \frac{dy}{dx} \right|_{x=2} &= \frac{7}{4} - \frac{84}{64} \\
 &= \frac{7}{16}
 \end{aligned}$$

When $x = 2, y = \frac{14}{4} = \frac{7}{2}$.

$$\begin{aligned}
 y - \frac{7}{2} &= -\frac{16}{7}(x - 2) \\
 14y - 49 &= -32x + 64 \\
 32x + 14y - 113 &= 0
 \end{aligned}$$

(b) When $x = 0, y = \frac{113}{14}$.**(c)** $d(3x^2 + 4) = 6x \, dx$

$$\begin{aligned}
 \int_0^2 \left(\frac{113 - 32x}{14} - \frac{7x}{\sqrt{3x^2 + 4}} \right) dx &= \left[\frac{113}{14}x - \frac{32x^2}{28} \right]_0^2 - \int_0^2 \frac{7}{6}(3x^2 + 4)^{-\frac{1}{2}} d(3x^2 + 4) \\
 &= \frac{113}{7} - \frac{32}{7} - \left[\frac{7}{6} \left(\frac{1}{2} \right) (3x^2 + 4)^{\frac{1}{2}} \right]_0^2 \\
 &= \frac{81}{7} - \frac{7}{12}(4 - 2) \\
 &= \frac{81}{7} - \frac{7}{6} \\
 &= \frac{437}{42}
 \end{aligned}$$